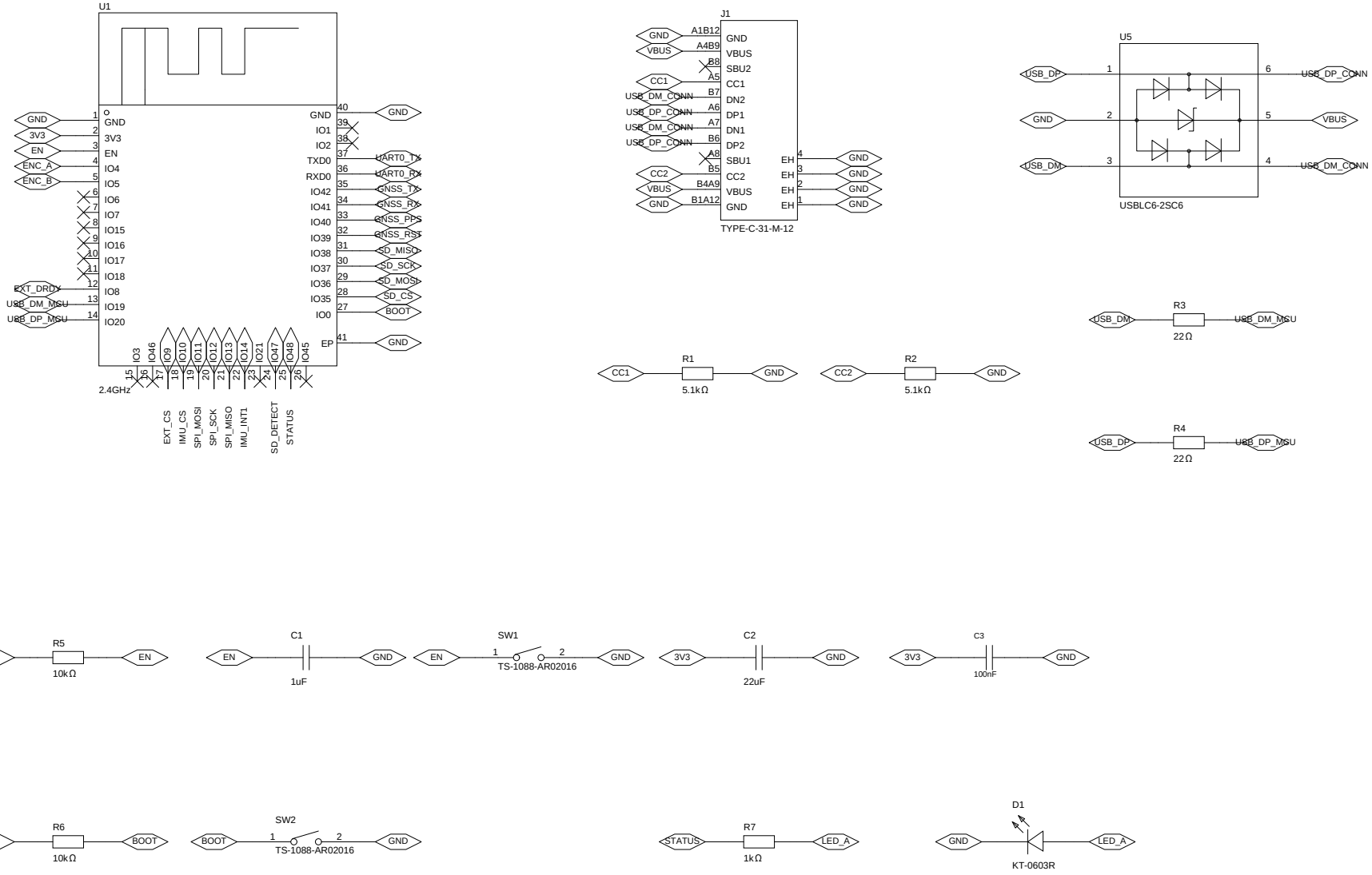


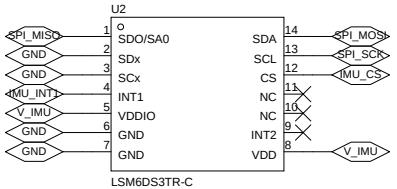

## 01 MCU / USB / BOOT - R2B

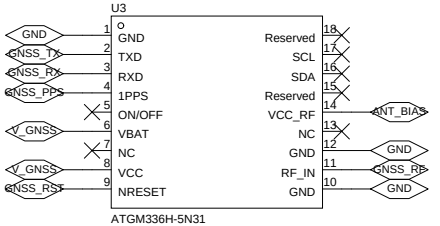
ESP32-S3-WROOM-1-N8 (no PSRAM); USB native FS

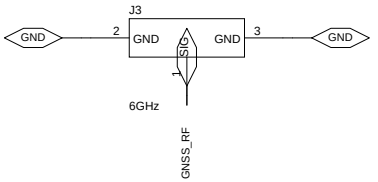
BOOT: hold SW2, tap SW1 RESET, release SW2

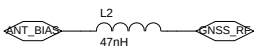
PCB bottom TP1=TX0 TP2=RX0 TP3=EN TP4=BOOT

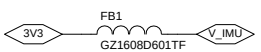
USBLC6: pins 1-6 and 3-4 connect internally; connector nets are split

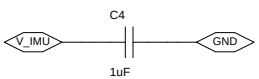


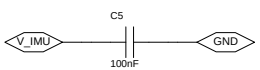



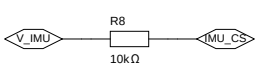



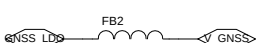



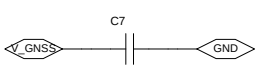



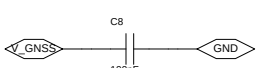






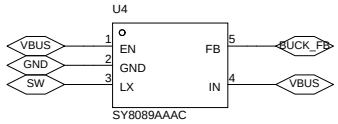



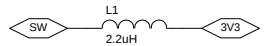

## 02 IMU / GNSS - R2B

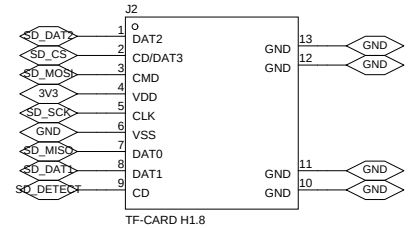
LSM6DS3TR-C SPI + INT1 / FIFO; ATGM336H-5N31 UART + PPS

J3: 3.3 V active antenna, prefer 20 mA or less; 50 mA is short-circuit limit

V_GNSS: dedicated 3.3 V LDO through FB2; VBAT shares supply

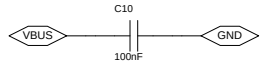


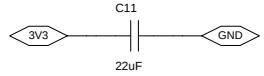



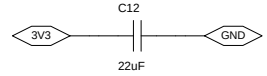



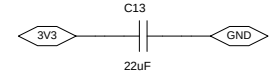


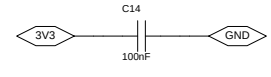


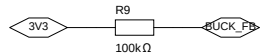


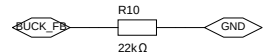



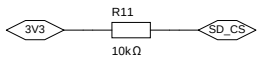



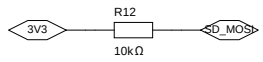



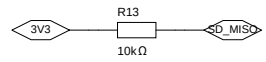



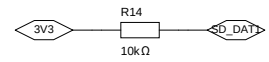



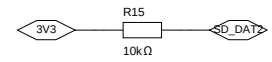



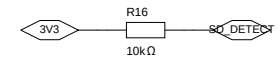



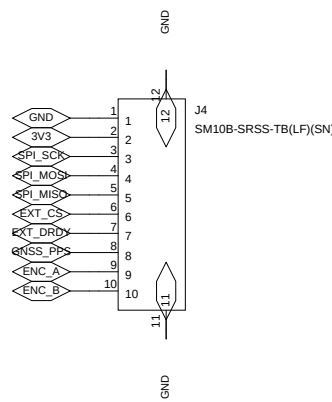



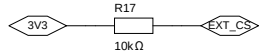

**03 USB POWER / microSD - R2B**

USB 5 V; buck output 3.327 V nominal; C9 now 22 uF

39 x 43 mm; four copper layers; 53 components / 25 part types, all top

5 assembled boards; TF card and antenna external

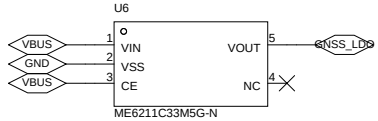


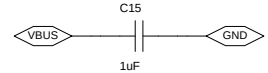



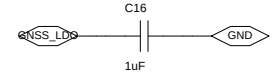

## 04 EXPANSION / 3.3 V LOGIC - R2B

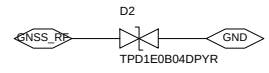
J4: 1 GND  2 3V3  3 SCK  4 MOSI  5 MISO

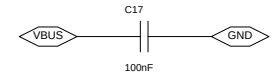
6 EXT_CS  7 EXT_DRDY  8 PPS  9 ENC_A  10 ENC_B

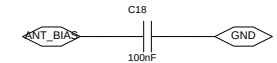



## 05 GNSS POWER / PROTECTION - R2B

U6: ME6211C33M5G-N; C15 input and C16 output before FB2

D2: TPD1E0B04DPYR (0.15 pF typ, 0.18 pF max), RF ESD shunt

C17: USBLC6 VBUS bypass; C18: 100 nF ANT_BIAS bypass (existing C14663 SKU)